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Technical Section

Evaluating user perception toward physics-adapted avatar in remote heterogeneous spaces ☆

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ABSTRACT

In avatar-mediated telepresence, it is essential to reflect the user's movements to the avatar for effective interaction. In the case of heterogeneous telepresence environments where two spaces differ in size and furniture layouts, researchers have proposed methods to adjust the avatar movement to deviate from the user's while preserving the intention to adapt to the physical environment. This study aims to investigate users' perception and preference toward their physics-adapted avatars in remote spaces. To this end, we conducted two user studies. In the first study, we examined how users perceive their avatars when their pose category is changed to avoid physics conflict. In the second study, we investigated users' perception when the avatar's eye level is modified to avoid physics conflict. The first study showed that the avatar's pose category modification led to a decrease in ownership and control over their avatar, decreased intention preservation and trust toward the system, and less preference. Physics conflict also negatively affected users' perceptions. Users' preferences for adapting their avatar's pose category to match the physics versus preserving the pose category, despite the physics conflict, were similar. The second study indicated that users recognized the hierarchy that occurred due to the eye level change, and preferred preserving their eye level. Based on these results, we offer suggestions for avatar adaptation in heterogeneous space telepresence systems.

1. Introduction

Avatar-mediated extended reality (XR) telepresence allows physically distant users to communicate as if they were in the same physical space. Early studies on telepresence did not consider the difference between two physical spaces either by dealing with scenarios that the spatial difference is irrelevant [1] or by assuming that the two nearby spaces around users have identical furniture layout [2]. Researchers have suggested different methods to overcome this problem. Some studies found mutually empty spaces such that users can interact only within free space while avoiding collision with the environment [3,4]. Other studies enabled the avatar to interact with the remote environment by placing it at a position corresponding to the users' location, either predefined based on furniture category [5,6] or calculated online by considering the user's context, such as the gaze and spatial relations with other objects [7]. Unconstraining the size of the avatar increases the freedom of locating the avatar in the AR physical space [8] or of navigating in a virtual space [9] for effective interaction. These methods showed impressive ways of improving telepresence systems by overcoming differences in space and system configuration.

To overcome the difference between two spaces, some of the above approaches adjust the avatar's motion to enable the avatar to naturally interact with the physical environment or to convey the user's context clearly [5,8,10,11], resulting in the discrepancy between user's and avatar's movement without being notified. We recognize the significance of improving the quality of interaction for local users to effectively engage with remote users by adjusting motions. However, we also put importance on avatars not deviating too much from the user without them being notified or to reduce uncanny distortions [12]. The remote users might not even need to recognize the distortion that takes place in their avatar since it is not happening within their view. However, might they actually care? If they do, to what extent? We were interested in exploring these questions.

We can think of an example as follows. Suppose you are talking to an avatar of a colleague working far away. You are sitting on a chair, talking about a research paper. At the same moment, your avatar is standing in the remote colleague's office since there is no corresponding chair that the avatar can sit on. You will not recognize the pose change since you cannot see nor access the remote space. Slater et al. [13] opened up the question of presence with "Where are you", which

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Fig. 1. One condition from our study 1 where the user's pose category is modified to fit the physical configuration(*i.e.* the existence of a chair) of the remote space. As a result, the avatar in the remote space is sitting. (A) is a third-person perspective view of the remote space. (B) is a third-person perspective view of the local space. User is observing their avatar using a portal. (C) is the user experiencing the condition.

emphasizes oneself immersed within the virtual environment when all sensory feelings are dominated through virtual reality (VR) systems. We ask another question, "Where is your avatar", that will appear in the remote space on your behalf. We should trust the avatar in the sense that, that other self will convey myself in the right manner. In this vein, there might be nuanced motion or action that needs attention when modified.

Specifically, this study aims to answer two questions. Our first research question is about the influence of modifying the pose category (*e.g.*, sitting and standing) and physics conflict (*e.g.*, penetrating into a physical object) on the user's perception of one's own avatar: **Are users willing to change their pose category if their avatar faces physics conflict in the remote space?** The second question pertains to the influence of modifying the user's eye level and the occurrence of physics conflict on the user's perception of the avatar: **Are users willing to change their eye level with the collaborating partner if their avatar faces physics conflict in the remote space?**

We conducted two user studies to answer two research questions raised above. For the first study, the user in the local space was either standing or sitting while interacting with the interviewer. However, the remote space might or might not have a chair in the position where the avatar should be placed, making it necessary to determine whether to preserve the user's pose category or not. Fig. 1 shows a snapshot of the first user study. For the second study, the user in the local space always looked straight at the interviewer, while the remote space includes furniture changes leading to a change in eye level between the avatar and the interviewer. Our experiment focused more on gauging users' general perceptions and preferences, with less emphasis on the feasibility of deployment in an augmented reality (AR) environment. Therefore, we conducted the user study in a VR environment which allowed us to overcome issues such as showing the other space with high resolution. The Portal, used to visualize the remote space (Fig. 1 B), was intended only for user study and not for actual AR telepresence applications. Dealing with mutual telepresence, we assume that two users send their avatars to remote space. For clarity, we will focus only on "the local space" with one user (referred to as "the user") and the user's avatar (referred to as "the avatar") located in a "remote space". The remote space has a collaborating partner user who has sent their avatar to the local space. We refer to that collaborating partner as "the interviewer" since our user experiment took an interview scenario.

In summary, our study investigated users' receptivity to modifying the motion of their avatars, which is crucial for the avatar adaptation approaches to be adopted by the user. Understanding how far avatars can deviate from the users to fit into dissimilar environments will enable us to take advantage of more spaces, acknowledging that complete alignment or spaces may not be achievable in all spaces. To the best of our knowledge, this is the first study to investigate users' perception and preference toward modified avatars of their own in a heterogeneous environment.

2. Related work

2.1. Heterogeneous telepresence environment

Researchers have developed various methods to realize collaborative mixed reality (MR) telepresence to bring remote people together into a single space [2,14–16]. Since users dwell in spaces with different configurations, researchers tackle this dissimilarity in diverse directions. The first approach focuses on optimal space transformation. Studies by [3,4] demonstrated that identifying and maximizing mutually empty spaces significantly improved the quality of telepresence interactions. Building on this, [5,6] found that strategic avatar placement based on local user positioning led to more natural and effective communication patterns.

The second approach invites remote users to the local user's space. This line of work reconstructs the local user's space into 3-dimensional (3D) AR space so that VR users can join the same space. Kumaravel et al. [17] revealed that combining view switching with holographic interfaces enhanced physical task collaboration between AR instructors and VR learners. Yu et al. [18] discovered that point cloud reconstructed avatars significantly outperformed virtual avatars in presence metrics, while Pakanen et al. [19] demonstrated a strong user preference for photorealistic full body avatars in both AR and VR settings. Piumsomboon et al. proposed Mini-Me [8], which changed the scale and orientation of the remote avatar to fit into the local users' field of view (FOV) to correctly translate the user's gaze and pointing direction. Their subsequent work [20] established the effectiveness of flying interfaces for large site collaboration. Yu et al. [21] proposed a framework that makes duplicate copies of the common region of interests to give individual users higher degrees of freedom for interaction. Notably, Numan et al. [22] uncovered distinct behavioral patterns: VR users exhibited greater physical movement, while AR users engaged in more verbal communication, though subjective experience ratings remained similar.

The third approach addresses a method to adjust the avatar's motion to fit the remote physical environment. Choi et al. [23] studied avatar locomotion and visibility for better representation in remote spaces and suggested a few rules when transcending local users' locomotion to the remote spaces. Yoon et al. [7] proposed a learning-based approach to place the avatar in a dissimilar remote space. Their method achieved re-positioning remote users to the local users' spaces regardless of remote users' locations. Wang et al. [24] suggested a "predict-and-drive" technique for a smoother transition of teleported avatars to overcome the delay that occurs when transcending remote avatars to the local spaces, Choi et al. [25] tackled retargeting the locomotion and object interaction motions in morphologically different spaces, and Kang et al. [26] tackled transmitting deictic gestures to remote spaces with different layouts to correctly convey the meaning of deictic directions. While these studies primarily addressed space heterogeneity through user positioning, motion adaptation, or space adaptation — or examined how the different configurations on other users' avatars affect the user's perception — our work takes another approach. We specifically investigate users' perceptions and preferences regarding their own avatar's presentation to others, rather than focusing on how remote avatars are perceived or modified.

2.2. Retargeting avatar motion

Several methods have been proposed to prevent physical collisions that create unnatural user movements caused by directly applying the user's movement from one space to another. Jo et al. [27] and Kim et al. [28] proposed motion adaptation in an AR environment, where the user's pose is transformed to fit into the physical configuration of another site. Kim et al. [29] suggested a method to preserve head gaze in heterogeneous environments with possible multiple users. Ullal et al. [30] proposed a method to overcome pose distortion due to changes in the interaction object. Some studies specifically focus on transferring deictic gestures. Sousa et al. [31] and Mayer et al. [32] warped pointing gestures to enhance interpretation. Piumsomboon et al. [8] and Ullal et al. [10] changed the direction of pointing in perceptible ways to fit within the local user's field of view or to preserve the pose. All these methods focused on enhancing the interpretation of gestures from the viewer's perspective. While understanding the importance of transmitting correct intention through retargeting, we further explore perceptions from a counter direction: how would the owner of the avatar perceive their transmitted self? Our study aims to explore users' perceptions and preferences toward retargeted avatars in heterogeneous spaces.

2.3. Perception of asynchronous representation

Since MR technology allows users to change their size, pose, appearance, and many other factors, researchers have studied users' perceptions toward revised versions of themselves. Much work focused on participants' perception of attached or extended body figures of virtual characters, such as the sense of ownership when a tail is attached [33], when using multi-hand systems [34], when using a mixture of multi-sensory stimuli [35], and when the size of body part changes [9]. These studies highlight that when the extended or attached parts align with or are affected by the user's movement control, the user's level of ownership and control still holds. Fribourg et al. [36] proposed "co-embodiment", where users share one virtual avatar. They found that users tend to significantly overestimate their sense of agency and perform similar body movements regardless of their actual level of control toward the avatar. Some focused on manipulating height in virtual reality to investigate social hierarchy perception and paranoia [37] or stature-related perceptions [38]. Both studies confirm that lowering or higher height affects the way users perceive themselves. Tao et al. [39] studied the effect of adding physics correction to self avatar motion to the user's embodiment that enhances the presence score. All of these studies focused on avatars in the same space as the users, meaning users recognize the change that happens to their avatars. Here, we investigate questions about users' remote avatars, situated in distant spaces on behalf of the real users. We focus on understanding users' perception of avatars sent to remote spaces when their pose category or eye level is asynchronously transferred.

3. Research question and hypothesis

We aim to examine whether users have an increased inclination to change their pose category or eye level when their avatar faces a physics conflict in a remote space. We have defined the hypotheses below.

- **H1-1: Not preserving the avatar's pose category will negatively affect all measures.** We expect that users will report a reduced Sense of Ownership (SoO), Sense of Agency (SoA), Trust, and Intention preservation toward the avatar, despite knowing that the avatar is not in their local space for direct interaction.
- **H1-2: When the avatar encounters physics conflict or collision in the remote space, the physicality perception will decrease.** We assume that the existence of physics conflict will negatively affect the physicality perception, which measures how much users felt physical realism, as well as Trust and Preference ratings. SoO, SoA, and Intention preservation might not differ significantly since only environmental factors changed between levels in physics conflict, not on avatar directly.
- **H1-3: When there is physics conflict in the remote space, the user's preference for not preserving the user's pose to the avatar will increase.** Studies (*e.g.*, [24,40]) suggest that reducing physical conflicts from the observer's perspective improves presence. Thus, we assume that users' inclination to change their pose category would increase in the face of physical conflict compared to the cases without physical conflict.
- **H2-1: There will be a significant negative effect on all measures when the user's eye level is not preserved.**
- **H2-2: When the avatar encounters physics conflict or collision in the remote space, users' physicality perception will decrease** for similar reasons as in H1-2.
- **H2-3: When there is physics conflict in the remote space, the user's preference for not preserving the user's eye level to the avatar will increase** for similar reasons as in H1-3.

4. Study 1: Pose category and physics conflict

The purpose of study 1 is to see the effect of pose category change when the avatar faces a physics conflict in remote space to the user. We chose the pose category for the following reasons. First, studies report an increased preference and embodiment when they have full control over their virtual body [41]. Also, one factor that affects the feeling of presence within the virtual environment is place illusion which happens through sensorimotor contingencies [42]. Lastly, body posture and motion are major factors in non-verbal behavior, and different postures can convey different moods [43–45]. In the following paragraphs, we refer to the participants as "users", except for Sections 4.3 and 4.4, to avoid confusion.

4.1. Study design

We examined the effects of three factors (two levels each) on user perception and preference: Factor 1: User's Current Pose Category (Current Pose), Factor 2: User's Pose Category Preservation (Pose Preservation), and Factor 3: Physics Conflict Existence (Physics Conflict). We refer to a pose as a particular static pose category. We chose the two most common pose categories when communicating with others: standing and sitting. For the Current Pose factor, participants either **Stand** or **Sit** while performing the task. For the Pose Preservation factor, we chose **Preserved (P)** and **Not Preserved (NP)**. In **P**, the user's full body tracking, including the pose category, was identically transferred to the avatar. In **NP**, we disabled lower body tracking and instead forced the lower body pose to either Stand or Sit by fixing leg joints' angles. We fixed the lower body to a specific pose, as identifying the adequate level of control over the lower body raised other questions. Thus, once the user's lower body was adapted, they lost control over it. Third, for the Conflict Existence factor, we chose **Exists (E)** and **Does not Exist (DE)**. In **Exists (E)**, we created an environment that is physically impossible in reality. For the **Stand** case, we created a visual collision between the avatar and a chair, such that the avatar penetrated the chair. For the **Sit** case, the avatar appeared as if it was floating above the ground without a chair. For **Does not Exist (DE)**, we designed an environment that does not violate the law of physics. For the **Stand** case, the avatar was properly standing on the ground without any chair, while for the **Sit** case, the avatar was properly sitting on the chair. These consist of two (Stand, Sit) x two (P, NP) x two (E, DE) factorial within-subject designs, as summarized in Fig. 3. An actor played the role of the interviewer in our experiment to provide consistent behavioral cues to the user. The first-person perspective (1PP) and third-person perspective (3PP) of each condition in the user experiment are shown in Fig. 2. 1PP views are slightly corrected to fix skewness for better visibility.

4.2. Study setup

This section describes how we implemented the heterogeneous telepresence setting and user/avatar body tracking. We used the Unity3D/SteamVR platform for system development and HTC Vive Pro devices [head-mounted display (HMD), two controllers, and three Vive trackers] for tracking and displaying the VR environment for the user experiment. We used characters both for the user and the interviewer from Ready Player Me [46] where we can customize the character's

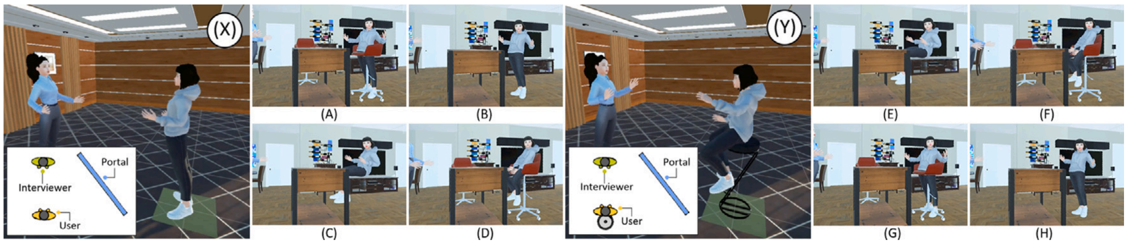

Fig. 2. (A)~(H) are the 1PP view of the user of eight study conditions of study 1. (X) and (Y) are two levels of the factor Current Pose displayed with 3PP and top view drawing. Note: these images might not be temporally aligned.

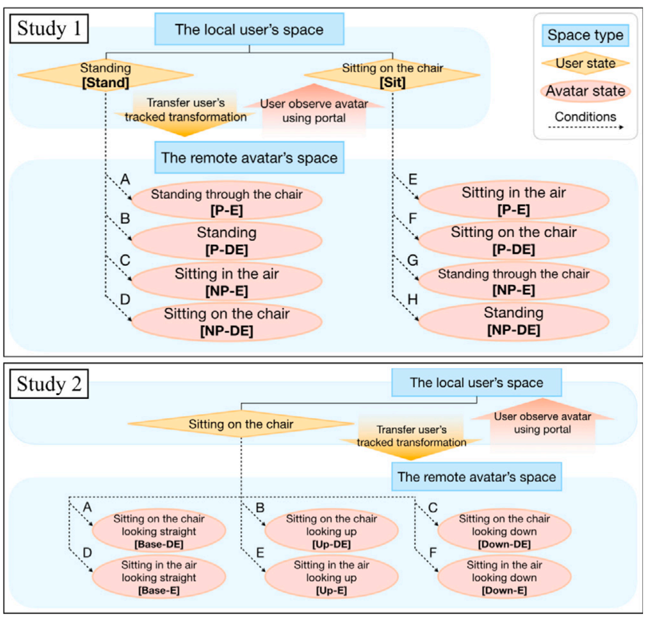

Fig. 3. Diagram of the study structures. Alphabet ordering corresponds to the same alphabet condition with Fig. 2 for study 1 and Fig. 5 for study 2.

look. We matched the gender of the avatar with that identified by the participant for the experiment.

Pilot Test

The purpose of the pilot test was to determine which method best visualizes the existence of the remote space and the modification of the avatars. The question we wanted to address was, through which portray, would the user understand the modification made to the avatar without being notified, simultaneously? Our focus was more on whether the user understands that the modification happens to their avatar, rather than assessing the user's perception toward the remote avatar. Within studies that focus on heterogeneous telepresence, there were very few works that display the remote space to the local users. One example was Kumaravel et al. [17] which captures local space in real-time and then transmits it to the remote space for collaboration. This copy-and-paste method was not directly applicable to our study since we have two local spaces where only "avatars" are transmitted. We considered several methods, including showing the remote space video afterward, changing perspective, and observing through the portal which streams how the avatar acts in the remote space simultaneously. We excluded post-hoc video showing because it could affect the Sense of Embodiment score due to embodiment differences since the experience and presence provided through video can differ from observing within a VR environment [47,48].

We conducted a pilot test comparing two methods: changing the perspective to the avatar and observing a portal. We focused on which visualization method can best show what is happening in the remote space. Changing the perspective lets users shift their view to the avatars' view, thereby users will feel as if they were transported to the remote heterogeneous spaces, to the avatar's body. Observing the portal is the method we used in our work which allows participants to observe the user and the avatar at the same time. We came up with this idea from [17] where remote space is replicated and partially fused in the local space simultaneously. Also, a portal is used in many scenarios to transport users from one place to another or like a window that shows the other place [49–51]. From this vein, the portal seems to suit our scenario where users have to view what is happening in remote spaces. The process of the pilot study was: participants first enter the scene and get adjusted to the calibrated avatar. Then, they started the task which was to answer questions and talk about their preferred food. After being asked two questions, participants were asked to compare the user and the avatar using each method. In changing perspective conditions, participants were teleported to the remote avatar's perspective to see the avatar. In portal condition, participants pop up the portal and see the user and the avatar at the same time. After each condition, participants filled out a survey. Survey questions consist of three questions focusing on which method makes it easier for participants to observe and understand the avatar's movements. (e.g., Using portal view or changing perspective makes it easy for me to compare "me" and "my avatar"). Responses were rated on a Likert Scale from one to seven and tested with eight participants. The mean and standard deviation for all three questions were similar between the two methods. We initially infer there would be one method that surpasses the other. However, similar statistical values were interpreted to us, such that participants recognized what was happening to their avatar and did not seem to be affected by the visualization method magnificently. We chose the portal as a visualization method because mainly, we wanted to reflect the real use case scenario that the user would not need to move into nor see the remote space. The portal view also allowed participants to directly compare the user and avatar without temporarily breaking out of their current avatar which will be "the user". Lastly, 3PP provided better space visibility to observe the avatar's pose and eye height changes compared to the 1PP [52].

Scene Description To demonstrate the concept of the heterogeneous environment, we constructed two spaces, one for the interviewer and one for the user, as shown in Fig. 1(A) and (B). The spaces had different interior styles to make them easily distinguishable. A desk was included between the user and the interviewer in the interviewer's space, but not in the user's space, to create the feeling of open space. The distance between the interviewer and the user was maintained in both spaces. We used a portal to display the remote space to the user. During the task, as in Section 4.3, Fig. 2, and Fig. 5, a user opened the portal to see how their avatar looks in the remote space. We placed a virtual camera in the interviewer's space to the side of the avatar to capture their full body movement and the desk in front of the avatar. The portal was positioned on the side on purpose, as the user did not need to see their avatar while speaking to the interviewer in the local space.

Local User We applied inverse kinematics (IK) to data from trackers worn by the user to enable full-body tracking. The HMD provides head input, two controllers track hand and arm movements, and three VIVE trackers attached to the pelvis and feet track the root and feet positions.

Users move around the virtual space using the controllers and trackers. An IK solver [53] calibrates the height of the user to match the height of the avatar and animates the virtual characters based on the tracked transformations.

Remote Avatar Some conditions make the remote avatar take a different pose category than the local user. As such, we had separate modules for upper-body tracking as well as full-body tracking. For conditions that required a preserved pose (A, B, E, F), the user's tracked data (HMD, two controllers, three Vive trackers) was directly transferred to the remote avatar. Then, the IK solver animated the avatar. For conditions that did not preserve the user's pose (C, D, G, H), the upper and lower body were generated differently. For upper body tracking, we used the information tracked from the HMD and controllers to apply IK. The lower body poses were fixed to maintain either the standing or sitting pose. The pelvis transformation was set to have the same height as the user's initial pelvis height (y value in Unity3D) and adjusted if the initial pelvis height and the sitting pelvis height were deviated too much.

Interviewer We used IK for the full-body movement of the interviewer based on trackers worn by the actor (HMD, two controllers, three Vive trackers) to both the remote (interviewer's space) and local (user's space) space for synchronous movement.

4.3. Procedure and task

The experiment procedure is summarized in Fig. 4. Upon arrival, participants provided demographic information and signed the consent form. We then explained the experiment, including the number of conditions, approximate time, and task. Participants were informed that an interviewer would ask job/research-related questions. Participants were then assisted in wearing an HMD, controllers, and trackers. Once participants were ready, the experiment began with the **Intro** scene. The purpose of the **Intro** scene was to familiarize participants with the scene settings, including the existence of the remote space and key definitions such as the user and the avatar, to prevent confusion during the questionnaire phase. Participants first calibrated themselves to the character when they entered the **Intro**. After calibration, we asked participants to imagine that they were physically present in the space as much as possible. Participants moved their arms and legs, and also moved around the space to be accustomed to full-body tracking. When the participants felt comfortable, the interviewer initiated a conversation by asking the participants to define "research/job" for themselves. After the participant answered, we guided them to open the portal and explained how they could observe their avatar in the remote space using the portal. The scene ended when the participant reported comfort with the environment.

Participants experienced nine scenes in total (Intro + eight conditions). We randomized the order of conditions for every participant to minimize the potential ordering effects. For the **Stand** cases, participants actually stood during the experiment. For the **Sit** cases, participants sat on a tall chair in the real environment, as they were sitting on a virtual tall chair in the VR environment. This was to help them imagine that they were physically located inside of the space. The task was questioning and answering between the interviewer and the participants. The topic of the question was research or job, depending on the participant's occupation. Previous studies utilized conversation tasks where participants discussed or negotiated specific topics [54]. We chose a research or job conversation task since (1) this conversation mimics real-world use cases for telepresence, (2) the topic provides structure while allowing natural variation, (3) the topic is not too personal yet engaging enough to make participants focus on interaction during the experiment and (4) the questions can be standardized across all conditions. Each scene began with the interviewer asking two questions and the participants' answers. During this period, participants did not know how their avatar looked since the portal was not visible. It was only after when participants finished answering two questions, the

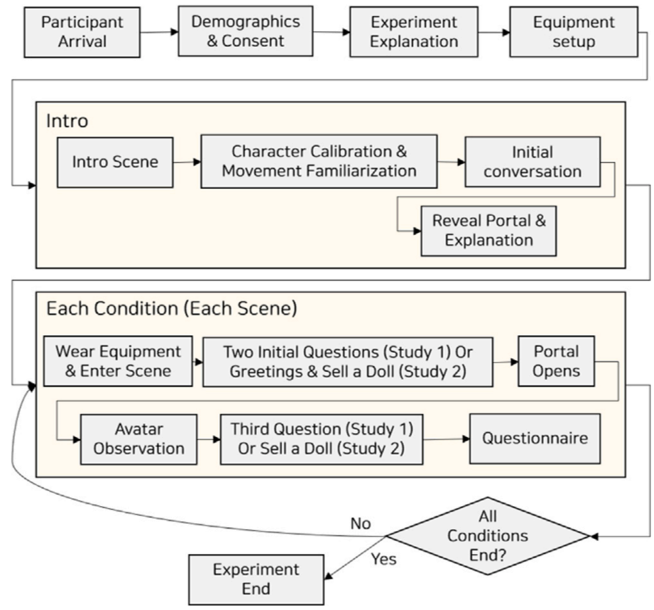

Fig. 4. Flow diagram of the experiment procedure for two user studies involving repeated question–answer sessions across conditions.

participants opened the portal and saw the avatar in the remote space. During the task, participants were instructed to stay within a green square not to generate dynamic motion such as walking. Participants observed both the user and avatar for as long as they needed. Last, the interviewer asked a third question and the participants answered. After experiencing each condition, except for the intro, participants filled out a questionnaire using a seven-point Likert scale. The total experiment procedure took approximately 90 min per participant. Regarding the questions, we used a single set of general questions for both participant groups. Questions included: (1) Please explain your major/job field. (2) How did you become interested in your field? (3) Who has had the most influence on your decision to pursue your current field of study/job? A total of 24 questions were given to each participant as there were eight conditions × three questions. These questions were prepared in advance and the order of the question was fixed.

4.4. Measures and participants

We measured six components using questionnaires adopted from previous studies, with modification of questions to focus on the perception of the avatar. Sense of Ownership (SoO) [55] identifies how much the user felt like the avatar was their body. Sense of Agency (SoA) [56] identifies how much the user felt the avatar's actions and consequences were under their control. Trust [57] identifies the user's trust level toward the system. Intention Preservation (Intention) identifies whether the user's intention, especially the body language, was correctly transferred to the avatar. Physicality [58] identifies whether the user feels their avatar seems to understand the laws of physics. Preference identifies the user's subjective preference toward the avatar's depiction. We used a seven-point Likert scale and added verbal words (*i.e.*, "Strongly disagree" to "Strongly agree") to help users rate their scores. Please refer to the appendix for the full questionnaire. An Institutional Review Board approved our study, and the number of participants was determined based on the related work [59]. We recruited 25 participants [10 male, 15 female; age range 20–30, mean(M) = 23.3, standard deviation(SD) = 2.39] via email and campus announcement boards and received written informed consent for participation. A total of 24 participants had prior experience with VR, while one had no experience.

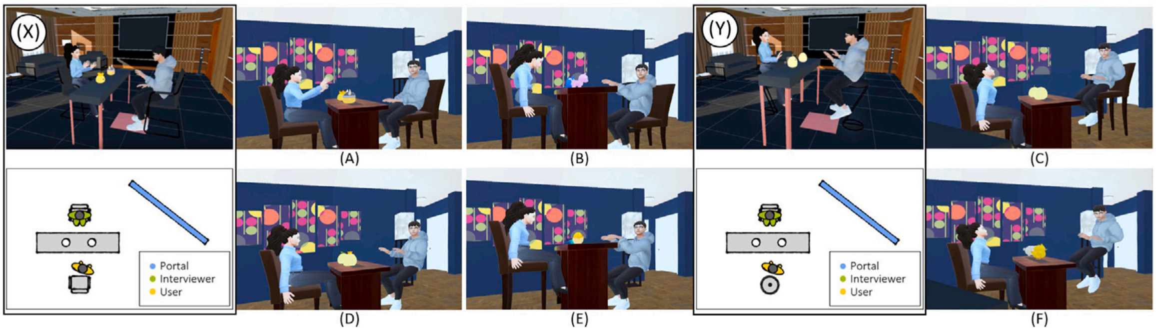

Fig. 5. (A)~(F) are the 1PP view of the user of six study conditions of study 2. (X) and (Y) are sitting status for the three levels of the factor Eye Level displayed with 3PP and top view drawing. Note: these images might not be temporally aligned.

5. Study 2: Eye level and physics conflict

Study 2 aims to see the effects of change in eye level with their collaborating partner when the avatar faces a physics conflict in a remote space. Here, eye level refers to the level at which two people talking face-to-face are positioned. We define two users as having the same eye level if they are sitting on chairs of similar heights and having different eye levels if one is sitting on a high-height chair (tall chair) and the other on an average-height chair. We chose eye level as a factor because it relates to social hierarchy and constitutes an important nonverbal behavior. Several studies have pointed out that greater height is often associated with greater dominance [60–62]. Burgoon et al. [63] have also noted that behaviors that increase height difference, such as "looking down at someone", are regarded as dominant behaviors with intention. Except for Sections 5.3 and 5.4, the term "the user" will be used continuously.

5.1. Study design

We examined the effects of Eye Level and Physics Conflict Existence on users' perception and preference of their avatar. For the Eye Level factor, there were three levels: Preserve (Base), Looking Up (Up), and Looking Down (Down). In **Base**, the eye level between the user and the interviewer in the local space is preserved in the remote space. In **Up**, the avatar looked up at the interviewer, while in **Down**, the avatar looked down at the interviewer. For the Conflict Existence factor, we formed a case that physically makes sense in **Does not Exist (DE)** and not in **Exists (E)**. In the DE case, the avatar sat on a chair. In the E case, the avatar floated in the air, as there was no chair. This experiment consisted of three (Base, Up, Down) x two (DE, E) factorial within-subject designs, as shown in Fig. 3. The 1PP and 3PP of each condition in the user experiment are shown in Fig. 5. 1PP views are slightly corrected to fix skewness for better visibility. As in study 1, an actor played the role of interviewer to provide consistent behavioral cues to the user.

5.2. Study setup

This section describes the detailed implementation of how we constructed a heterogeneous telepresence setting, as well as user/avatar body tracking. We used the same software and hardware system as in study 1, but the scene interior and avatar modification differed.

Scene Description We created two virtual spaces for a heterogeneous environment: one for the interviewer and one for the user with a different interior style. To analyze the effect of eye level difference in study 2, we manipulated the furniture height of the local and remote spaces to create a height difference between conditions. In the Base level, both the user and the interviewer sat on chairs and desks of an average height commonly used in office settings, in both local and remote spaces. In the Up and Down levels, the furniture heights differed

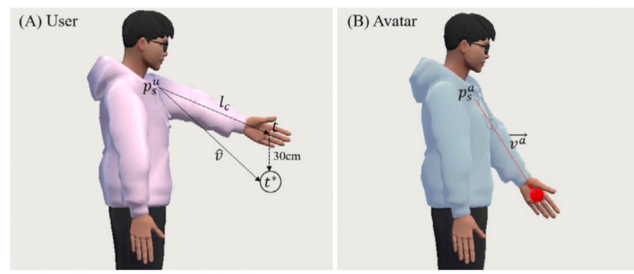

Fig. 6. Adjusting the avatar arm pose when the desk in the avatar space is lower than that in the user space. The avatar's target hand position t^* is set 30 cm lower than the user's hand position t. Then the avatar's arm direction $\vec{v}^a$ is rotated to point toward t^*.

between local and remote spaces. In the Up level, the local space had a chair and desk of an average height, with the user looking straight at the interviewer who was also sitting on a chair of average height. In the remote space, however, the interviewer sat on a tall chair and a tall desk while the avatar sat on an average-height chair, causing the avatar to look up. In the Down level, the local space had a tall chair and tall desk, with the user looking straight at the interviewer who was also sitting on a tall chair. However, the remote space had a chair and desk of an average height for the interviewer but a tall chair for the avatar, making the avatar look down. The height difference between the tall desk and the average desk was set to 30 cm, later used to calculate the warping angle for two arms. We changed the chair and desk height of the physical space during the experiment to match the virtual furniture. The distance between the interviewer and the user in the local space was preserved in the remote space. We used the portal to display the remote space to the user, located slightly further back compared to the previous experiment to show the eye level difference between the avatar and the interviewer.

Local User We used the same method as study 1. Tracked transformations from six devices (HMD, two controllers, and three Vive trackers) were used for IK in real-time.

Remote Avatar In study 2, we warped the avatar's head and arms to account for the height differences between the user and the avatar in the Up and Down conditions. To preserve the user's pointing intention in the avatar's space, we adjusted the arm movement accordingly. When the user pointed toward an object in front of them, the avatar's pointing target was set either 30 cm above (in Up cases) or 30 cm below (in Down cases).

To compensate for the difference between the user and the avatar target, we assumed that the hand location of the avatar was 30 cm above or below the user. We then applied IK to the new location. Fig. 6 illustrates this process.

We also adjusted the user's head movement to maintain eye gaze direction. Eye gaze direction in this context refers to the eye level and direction between the user and interviewer in a local space. Because the chair height might be different between the user and interviewer, the sitting height in local space might not match the one in remote space. To address this issue, we warped the user's head by projecting a ray from the HMD in the frontal direction to the height of objects on both tall and average desks. The two rays created an angle difference when projected to the two-dimensional plane, and applied to the avatar's HMD's rotation. The resulting angle varied depending on the person but was generally between 15 to 20 degrees.

Interviewer We utilized upper-body tracking for the interviewer, as they remained seated in both spaces using data from HMD and controllers. The interviewer's head and arms were adjusted with the same method used for the user to compensate for height differences.

5.3. Procedure and task

The experimental procedure is summarized in Fig. 4. Study 2 followed the same procedure as study 1 until the "Intro" scene. The task was changed to a role-play where participants acted as sellers and persuaded an interviewer to buy objects. We selected this sales-based role-play task for reasons similar to those in study 1: it facilitated natural, focused conversation while ensuring consistency in questions across all conditions. Participants completed a total of seven scenes (Intro + six conditions). We randomized the order of conditions for each participant. Throughout all conditions, participants were seated on a physical chair that corresponded to a virtual chair in the VR environment, with a real desk positioned to match the height of its virtual counterpart. Depending on the condition, participants sat on a tall or average-height chair and observed a tall or average-height desk.

The interviewer started by welcoming remarks. Next, the participant was asked to sell a doll on the left from the participant by directly answering the following question: "Please describe the doll on the left from your perspective and persuade me why I should buy this doll". After the participant answered, they saw their avatar in a remote space by popping up a portal. Participants observed the portal for as much time as they needed. Then, the interviewer asked the participant to sell the object on the right. The participant remained seated in a fixed position during the task. The interviewer followed a standardized protocol by accepting both offers. After each condition, the participant filled out a questionnaire on a seven-point Likert scale. After the end of all conditions, we conducted a short interview. The entire experiment took around 90 min per participant.

5.4. Measures and participants

We measured Hierarchy [64], which assesses whether participants perceive a difference in social roles between themselves and their avatars, along with study 1's metrics. We used a seven-point Likert scale with verbal anchors (Strongly disagree ... Strongly agree) to rate the scores. Our study was approved by an Institutional Review Board. We recruited 28 participants through emails and campus announcement boards, 18 of whom identified as male and 10 as female, with ages ranging from 20 to 30 (M = 25.21, SD = 2.68). All participants had experience with VR: 18 had 1–10 times of experience, seven had 11–50 times of experience, and three had heavy experience. 12 participants had experience with VR collaboration, while 16 had none. The sample size was set based on the related work [59].

6. Results

6.1. Study 1

We ran the Shapiro–Wilk test to data in all conditions and the tests revealed that the assumption of normality was violated in several conditions ($p < 0.05$) across all measures. We applied Aligned Rank Transform (ART) as proposed by Wobbrock et al. [65] for non-parametric factorial analysis of variance (ANOVA) to handle three independent variables. For measures that showed the main effect, we used the nonparametric Wilcoxon signed-rank tests as a post-hoc analysis to analyze the effect of each factor. We include interpretations of effect sizes using rank-biserial correlation (r_B) according to the following [66]. We used non-parametric statistical methods for post-hoc analysis since they tend to be more conservative by violating fewer assumptions [67,68]. We applied the Holm correction to control for multiple comparisons in our post-hoc tests. We report the summarized results of the factors that showed the main effect as illustrated in Fig. 7. Current Pose did not show significance on any measure, while the other two factors showed significant differences between levels.

6.1.1. Sense of ownership (SoO)

Factor Current Pose did not show significance ($p = 0.354$). Pose Preservation had a significant main effect $F(1, 168) = 37.239$, $p < 0.001$, $\eta_p^2 = 0.409$. Post-hoc analysis revealed that participants SoO was higher in P ($M = 4.37$, $SD = 0.90$) than in NP ($M = 3.75$, $SD = 0.90$) [$z = 5.278$, $p < 0.001$, $r_B = 0.637$ (large)]. Physics Conflict showed a significant main effect $F(1, 168) = 7.433$, $p < 0.007$, $\eta_p^2 = 0.129$. SoO was higher in DE ($M = 4.21$, $SD = 0.94$) compared to E ($M = 3.91$, $SD = 0.95$) [$z = 3.242$, $p < 0.001$, $r_B = 0.396$ (medium–large)],

6.1.2. Sense of agency (SoA)

Factor Current Pose did not show significance ($p = 0.701$). Pose Preservation had a significant main effect $F(1, 168) = 5.089$, $p < 0.001$, $\eta_p^2 = 0.476$: SoA increased in P ($M = 4.10$, $SD = 0.60$) than NP ($M = 3.61$, $SD = 0.67$) [$z = 5.619$, $p < 0.001$, $r_B = 0.686$ (large)]. For Physics conflict, Factor Physics Conflict did not show a significant main effect $p = 0.360$.

6.1.3. Trust

Factor Current Pose did not show significance ($p = 0.383$). Pose Preservation had a significant main effect $F(1, 168) = 54.710$, $p < 0.001$, $\eta_p^2 = 0.383$. Trust increased in P ($M = 4.42$, $SD = 0.80$) compared to NP ($M = 3.66$, $SD = 1.03$) [$z = 6.015$, $p < 0.001$, $r_B = 0.711$ (large)]. Physics Conflict showed a significant main effect $F(1, 168) = 23.284$, $p < 0.001$, $\eta_p^2 = 0.225$. Trust increased in DE ($M = 4.29$, $SD = 0.97$), compared to the E ($M = 3.80$, $SD = 0.98$) [$z = 4.500$, $p < 0.001$, $r_B = 0.565$ (large)].

6.1.4. Intention

Factor Current Pose did not show significance ($p = 0.487$). Pose Preservation had a significant main effect $F(1, 168) = 113.159$, $p < 0.001$, $\eta_p^2 = 0.467$: intention with a score of ($M = 5.42$, $SD = 1.11$) in P than in NP ($M = 3.90$, $SD = 1.48$) [$z = 7.16$, $p < 0.001$, $r_B = 0.890$ (large)]. Physics Conflict also showed a significant main effect $F(1, 168) = 5.715$, $p < 0.017$, $\eta_p^2 = 0.057$. Intention preservation perception was high in DE ($M = 4.87$, $SD = 1.54$), compared to E ($M = 4.46$, $SD = 1.47$) [$z = 3.011$, $p < 0.002$, $r_B = 0.411$ (medium–large)].

6.1.5. Physicality

Factor Current Pose did not show significance ($p = 0.857$). Pose Preservation had a significant main effect $F(1, 168) = 56.052$, $p < 0.001$, $\eta_p^2 = 0.306$. Physicality was higher in P ($M = 4.37$, $SD = 0.86$) compared to NP ($M = 3.91$, $SD = 0.90$) [$z = 4.357$, $p < 0.001$, $r = r_B = 0.554$ (large)]. Physics Conflict showed a significant main effect $F(1, 168) = 29.202$, $p < 0.001$, $\eta_p^2 = 0.331$. Physicality increased in DE ($M = 4.39$, $SD = 0.91$), compared to the E ($M = 3.90$, $SD = 0.84$) [$z = 4.875$, $p < 0.001$, $r_B = 0.631$ (large)].

6.1.6. Preference

Factor Current Pose did not show significance ($p = 0.629$). Pose Preservation had a significant main effect $F(1, 168) = 57.723$, $p < 0.001$, $\eta_p^2 = 0.334$. Participants reported a higher preference for cases when their pose was preserved ($M = 4.74$, $SD = 1.50$) compared to when it was not preserved ($M = 3.38$, $SD = 1.55$) [$z = 6.289$, $p < 0.001$, $r_B = 0.759$ (large)]. Physics Conflict showed a significant main effect $F(1, 168) = 50.505$, $p < 0.001$, $\eta_p^2 = 0.311$. Participants reported higher preference in DE ($M = 4.69$, $SD = 1.55$), compared to E ($M = 3.43$, $SD = 1.55$) [$z = 6.108$, $p < 0.001$, $r_B = 0.746$ (large)].

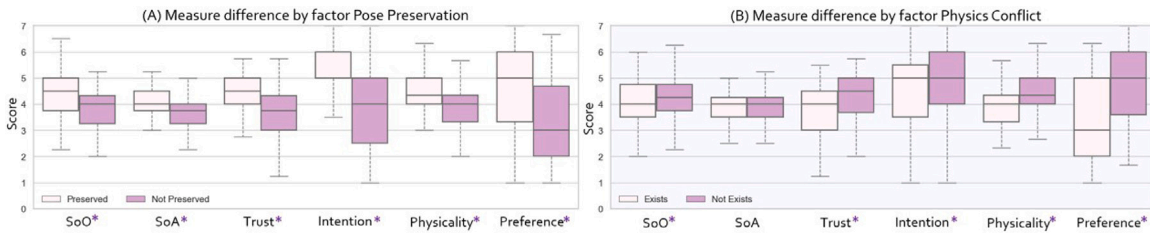

Fig. 7. Statistical results for all measures by the factor Pose Preserve (A) and Physics Conflict (B) of study 1. Significance is marked with an asterisk (*).

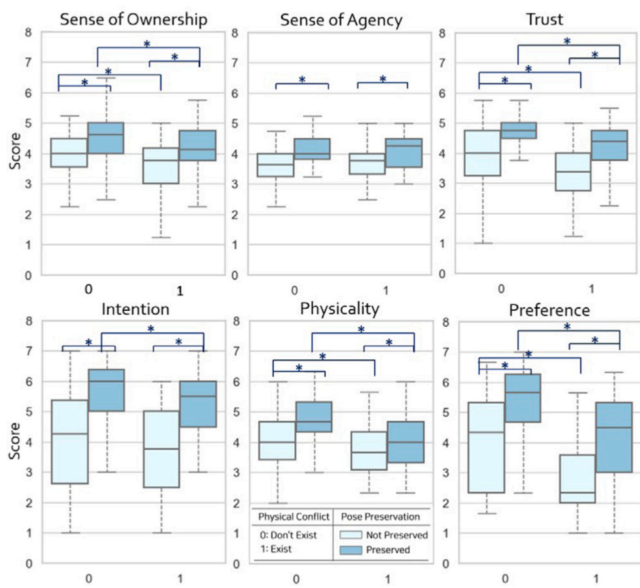

Fig. 8. Statistical results of pairwise comparisons between conditions for all measures in study 1. Significance is marked with an asterisk (*).

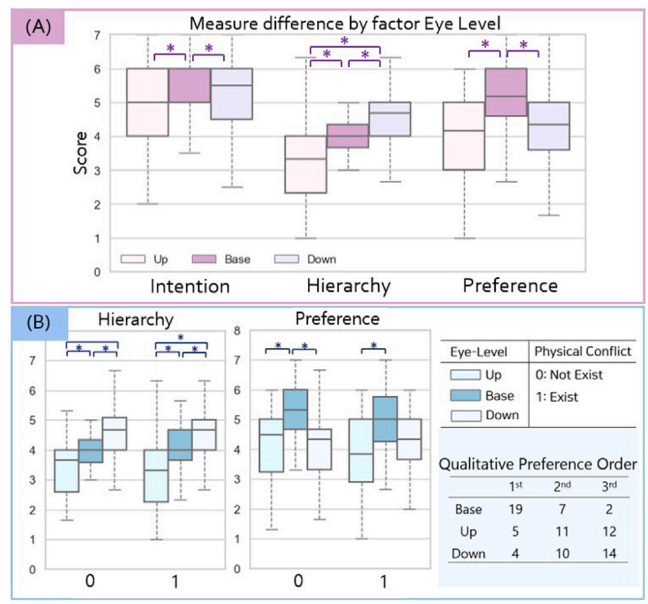

Fig. 9. (A) Statistical results for measures by the Eye Level of study 2. (B) Statistical results of pairwise comparisons between conditions for measures in study 2. Significance is marked with an asterisk (*).

6.1.7. Pairwise comparisons between conditions

Across all measures, no significant interaction effects were found between the factors (all $p > 0.05$). However, we analyzed the pairwise comparison to examine the trend as an experimental analysis. For pairwise comparisons between conditions, we first collapsed the data across Current Pose since it did not show any significance. Then we ran the nonparametric Wilcoxon signed-rank tests to compare each condition pair. The result statistics are summarized in Fig. 8. In general, conditions with an identical environment to that of the local space (P-DE) generally showed the highest score. For Preference, the condition in which the pose was preserved without physical conflict (P-DE) ranked the highest, whereas the condition without pose preservation but with physics conflict (NP-E) ranked the lowest. The mean difference between P ($M = 5.38$, $SD = 1.23$) and NP ($M = 4.10$, $SD = 1.49$) within the DE group (M difference = 1.28) and the mean difference between P ($M = 2.76$, $SD = 1.32$) and NP ($M = 4.00$, $SD = 1.53$) within the E group (M difference = 1.24) were not significantly different.

6.2. Study 2

A Shapiro–Wilk test indicated that the data were not normally distributed to several measures. We applied Aligned Rank Transform (ART) [65] like study 1 for non-parametric factorial analysis of variance (ANOVA) to maintain a consistent analytical framework. Significant main effects were found for Intention, Hierarchy, and Preference. As these measures did not meet normality assumptions, for post-hoc analysis, we applied the nonparametric Friedman's tests for the factor ***Eye level*** (3 levels) followed by post-hoc Wilcoxon signed-rank tests for those that showed significance, and Wilcoxon signed-rank tests for ***Physics Conflict*** (2 levels). Statistical results are summarized in Fig. 9(A) with an asterisk (*) that showed significance. Then we analyzed pairwise comparisons between conditions as in Fig. 9(B).

6.2.1. Sense of ownership (SoO), sense of agency (SoA), trust

For SoO, neither Eye Level nor Physics Conflict had a significant main effect with ($p = 0.164$), and ($p = 0.346$) respectively. SoA also did not show a significant main effect on either Eye Level or Physics Conflict with ($p = 0.124$), and ($p = 0.484$) respectively. For Trust, neither Eye Level nor Physics Conflict had a significant main effect with ($p = 0.085$), and ($p = 0.428$) respectively.

6.2.2. Intention

Eye Level had a significant main effect $F(2, 135) = 5.891$, $p < 0.003$, $\eta_p^2 = 0.495$. Intention was highest for the Base ($M = 5.59$, $SD = 0.99$) compared to Up ($M = 4.96$, $SD = 1.29$) or Down ($M = 5.14$, $SD = 1.11$) [$\chi^2 = 9.591$, $p < 0.015$, $W = 0.086$]. Factor Physics Conflict did not show a significant main effect ($p = 0.686$).

6.2.3. Physicality

For Physicality, neither Eye Level nor Physics Conflict had a significant main effect with ($p = 0.280$), and ($p = 0.299$) respectively.

6.2.4. Hierarchy

Eye Level had a significant main effect $F(2, 135) = 3.009$, $p < 0.001$, $\eta_p^2 = 0.499$. For the measure Hierarchy, a higher score is interpreted as participants feeling higher social hierarchy [$\chi^2 = 42.511$, $p < 0.001$,

$W = 0.380$]. This subjective feeling was reported to be highest in the Down ($M = 4.55$, $SD = 0.96$), followed by the Base ($M = 4.01$, $SD = 0.67$) and the Up ($M = 3.26$, $SD = 1.08$). Hierarchy showed significance between all pairs, Base-Up [$z = 4.148$, $p < 0.001$, $r_B = 0.695$ (large)], Base-Down [$z = -3.570$, $p < 0.001$, $r_B = -0.640$ (large)], and Up–Down [$z = -5.342$, $p < 0.001$, $r_B = -0.877$ (large)]. Factor Physics Conflict did not show a significant main effect ($p = 0.996$).

6.2.5. Preference

Eye Level had a significant main effect $F(2, 135) = 14.213$, $p < 0.001$, $\eta_p^2 = 0.495$. Participants reported the highest preference toward the Base ($M = 5.05$, $SD = 1.24$). The mean preference value for the Down ($M = 4.27$, $SD = 1.15$) was higher than the Up ($M = 3.98$, $SD = 1.35$) [$\chi^2 = 21.176$, $p < 0.001$, $W = 0.189$]. Post-hoc tests showed that the pair Base-Up [$z = 4.034$, $p < 0.001$, $r_B = 0.662$ (large)] and Base-Down had significance [$z = 3.349$, $p < 0.001$, $r_B = 0.555$ (large)], while Up–Down did not ($p < 0.166$). Factor Physics Conflict did not show a significant main effect ($p = 0.551$).

6.2.6. Pairwise comparisons between conditions

As in study 1, we analyzed the pairwise comparison to examine the trend as an experimental analysis since no significant interaction effects were found between the factors across all measures (all $p > 0.05$). We ran the nonparametric Wilcoxon signed-rank tests to compare the effects of the two factors in each pair of conditions. Part of the statistics is summarized in Fig. 9(B) with an asterisk (*) between conditions that showed significance. We found significance between Base-Up and Base-Down when physics conflict did not exist (DE) for Intention, and Preference. For Hierarchy, we found significance between every pair. In measure Preference, Base generally showed the highest score both in level DE (Base ($M = 5.27$, $SD = 1.09$), Up ($M = 4.06$, $SD = 1.34$), Down ($M = 4.17$, $SD = 1.22$)) and E (Base ($M = 4.82$, $SD = 1.36$), Up ($M = 3.91$, $SD = 1.39$), Down ($M = 4.37$, $SD = 1.09$)).

6.2.7. Qualitative comments

At the end of the experiment, we asked participants three questions. The first question asked was to identify their direct preference between conditions with physics impossibility. Fig. 9(B) shows their preference order from three given conditions (B, C, D). Second, we asked for their opinions on the good and bad parts of their avatar depiction. Third, their interest in knowing about the possibility of avatar adaptation to fit the remote space was asked, even if they could not see or did not need to in reality. These comments will further be discussed in Section 7.2.

7. Discussion

Our experiment aimed to examine the relative differences between conditions, with less emphasis on evaluating the absolute levels of the system usability. In a usability experiment evaluating a novel interaction method, a neutral score might be challenging to interpret, as it could reflect difficulties in discerning user preferences. However, given the within-subject design of our experiment, subtle differences in scores can reflect consistent trends in participants' experiences across conditions. While our results do not allow us to suggest an absolute threshold to what is allowed or not, our study detects these nuanced differences in individual responses to varying conditions. From this perspective, we describe how our result aligns with previous research and give further insights into the physics-adapted avatar in remote spaces.

7.1. Analysis on study 1

Preserving the user's pose was found to be better than not preserving it in all measures. This resistance seems to align with the place illusion suggested by Slater et al. [42], where the sense of being can be affected by sensorimotor contingencies. Other research also emphasized a higher sense of presence when there is a contingency between the virtual avatar [69]. Previous studies have explored asynchronous user representations [36,39], demonstrating that a sense of presence and control can be maintained despite certain discrepancies. These investigations typically allow users to observe avatar modifications within the same space the user stays. Our study wants to ask more questions adding to these existing works. What would be the user's perceptions and preferences toward the avatar that is not necessarily visible to them but will be seen by the other person in the remote space? Here, users might not be initially aware of avatar pose modifications. As an answer, we found that users may report diminished feelings of ownership, control, and overall preference. For the pose inconsistency, we observed many users giving similar comments about assuring whether what they are observing through the portal is not a "system flaw". When users first saw their avatar, they asked questions like "I am sitting...is this correct?"(P12) or "I think there is a flaw...could you check?"(P15) Although the operator did not answer these questions, these perceptions seem to affect the overall score. The unexpected motion out of control seems to decrease the credibility and affect the presence score. This finding potentially introduces nuanced considerations for avatar pose modification. Therefore, we accept H1-1.

Physicality was higher when physics conflict did not exist, supporting H1-2. We assumed that there would be no difference in the sense of ownership (SoO), control (SoA), and intention between DE and E. That was because SoO, SoA, and intention mainly ask users' perceptions toward the avatar, not the environmental differences. Contrary to our belief, SoO and Intention turned out to be higher in DE, while SoA did not show a significant difference. It is reported that merely observing the body discontinuity decreases the ownership [70], and physical collision affects the sense of ownership [71]. Our result seems to align with previous works in that users felt a higher sense of ownership toward the avatar's body when there was no physics conflict, but the degree of control remained unclear or similar. Trust in the system was lower when physics conflict existed in the avatar's space. Participants easily recognize the difference between the local space and the remote space, inferred from comments like "Oh, I am penetrating the object"(P8), and "the other space looks different"(P16). This recognition seems to affect the measure of trust since participants in general, seemed to assume that the avatar will not face a physics conflict just like the local space.

We analyze the measures that showed the main effects using nonparametric tests by factors as above that showed a clear significance and continue the discussion with the trends found between pairwise comparisons between collapsed conditions as in Fig. 8. Although there was no interaction effect, we analyzed the trend between conditions. Mean differences of the measure Preference between P and NP in DE and E were not significantly different, suggesting that users strongly prefer preserving their pose category, regardless of whether it involves physics impossibility. This strong preference seems to prevent an interaction effect between factors. Additionally, users showed a similar preference for condition P-E ($M = 4.10$, $SD = 1.49$) and condition NP-DE ($M = 4.00$, $SD = 1.53$). We assumed that users would show an increased inclination to change their pose category when there is physics conflict for the reason mentioned in Section 3: avoiding physics impossibility is best from the observer's point of view [24,40]. However, the result did not support our hypothesis. Therefore, we reject H1-3.

7.2. Analysis on study 2

We found that preserving eye level was better than the other two cases for intention, but its impact on SoA was not significant. Intention asks whether the user's intended body language was correctly transferred to the avatar. We could deduct the reason behind this result through the user's comments which later also seems to affect the preference scoring. It seems like users generally felt the change of eye level changed their original intention toward the counterpart. Participants in study 2 expressed opinions during post-hoc interviews about their will to not convey unintended social cues or behavior by mentioning words like: "It would be weird if I was talking in this pose, looking straight to the interviewer, in my thoughts ... but it turned out that it wasn't from the other person's eyes."(P8), "I was looking straight to the partner in my space but I was not in the remote space. I did not like that feeling."(P6), "Looking down other person looks uncomfortable"(P11). Hierarchy appeared in order of visible hierarchy, meaning that the users recognized the position change between the local and the remote space. Preference was highest in Base, implying that users preferred to preserve the local social environment in remote spaces. With this analysis, we partially support H2-1.

Unlike pose change, there were no conditions with a direct physical collision between the avatar and chair. Avatars sit in the air in scenes with physical conflict. Instead, the eye level difference seems to be more recognized by the user, showing more statistical significance as in Fig. 9(A). Since Physicality did not show either the main effect or the interaction effect, we reject H2-2.

Looking at pairwise comparison, for preference, the Base-Up mean difference in E (M difference = 0.92) was smaller than in DE (M difference = 1.21). Similarly, the Base-Down mean difference in E (M difference = 0.46) was smaller than in DE (M difference = 1.10). Intention also showed a smaller effect of eye level changes in E than in DE. From these observations, we interpret that there is a tendency to allow more eye level changes when there is a physics conflict. Therefore, we partially accept H2-3. Although the tendency to modify the eye level slightly increased, users seem to rate higher on preserving the eye level despite the physics conflict. The mean of condition Base-DE was higher than Up-E and Down-E, and we found a similar trend in the post-experiment interview. Participants seem to focus mostly on preserving the eye level of the local space to the remote space. Users stated that "preserving eye level will allow comfortable talking environment" (P3), "preserving eye level seems fair"(P17), and it is more "general"(P27). Post-hoc tests did not show a significant preference between the Up and Down. The number of participants who chose Up or Down in the interview did not differ significantly as well. We speculate that the factor hierarchy played a different role depending on the participants. For example, users who chose B over C mentioned, "I don't want to be high in the social hierarchy considering this is an interview situation"(P4), and "Looking down looks uncomfortable"(P11). Those who chose C over B also mentioned hierarchy, but with different contexts like, "I might look too submissive"(P17), and "I feel like my hierarchy is too low when my avatar is down"(P26).

For study 2, we conducted a semi-structured interview to identify whether participants wanted to know about the remote avatar adaptation and, if so, why, and if not, why not. About two-thirds (18) of participants wanted to know about avatar adaptation, Users who wanted to know directly mentioned comments related to how they will look to others like, "How I look to others is important, so I want to see it"(P18), "It is really important to transfer correct body language"(P5), and "Checking my current status will have either a negative or positive effect on me (so I want to know)"(P16). They also recognized a sense of the difference between themselves and their avatar and reported "a sense of difference", which sometimes developed into dissatisfaction. At the same time, users also showed concerns regarding the portal as having too much information, leading to a preference for having it as an option to turn on and off.

7.3. General analysis and suggestions

Although the number of participants was not exactly divided by the number of conditions, we observed that p-value significance was already clear with a smaller number of participants in both study 1 and study 2. So we aimed to maximize the number of participants. Comparing the two experiments, we understand the direct transformation of their pose category, which is more noticeable by the user is less preferred by the participants. They showed more resistance when their pose was changed compared to the eye level change. During study 1, participants frequently asked whether the system was performing correctly when they saw their modified avatar. However, in study 2, participants seem more tolerant toward changes. Statistical results are in line with their verbal response.

Compared to existing works, our work changes pose category or eye level as a means to save physicality, while other works suggested pose changes to enhance interaction quality. However, existing pose changing methods have not explored user preferences toward their modified avatar in the remote space that can deviate. Therefore, it would be interesting to investigate the results in terms of user acceptability when comparing pose changes to interaction quality. For example, work by Fidalgo et al. [72] reports higher co-presence with slight distortion of gestures. Recognizing that motion modifications are inevitable in some cases due to FoV [8], or to fit the diverse real environments [26], which can include highly different room and furniture layouts, the preference for pose change may vary depending on the task at hand and its contribution to the interaction quality.

From our initial result, our work can be extended to include more factors that can affect users' preference toward avatar modification: putting participants into more non-formal environments, more poses, two-party interaction, and mixed devices. Rendering styles such as using more photorealistic avatars and scenes might have another effect as Yu et al. report that the reconstruction-based looks and approaches are affected by task types [18]. As mentioned earlier, it may also be valuable to analyze the preference for pose changes to enhance interaction quality. In this study's scope, it was challenging to create experimental conditions with more diversity while ensuring the independence of each condition. Our study designs already required participants to experience eight or six different conditions. Adding more factors and levels would have increased the number of conditions and experiment time dramatically. However, we believe understanding how far avatars can deviate from the users to fit into dissimilar environments, would help us to adapt and leverage a wider range of environments, acknowledging that not all spaces can be perfectly aligned.

With these analyses, we propose a few suggestions when transferring a user to an avatar in a heterogeneous telepresence system. First, changing the user's pose category should be carefully designed when needed especially to simply fit the physicality. Second, it is recommended to avoid physics conflict for the avatar, even in remote spaces. Third, when collaborating with others, it is preferable to maintain eye level between the user and the others in the local space to the remote space, especially when its height is relatively equal. Finally, we suggest providing users with the option to check their avatar when desired, as many participants reported wanting to know their remote condition.

8. Limitations, future work, and conclusion

Although we made the first attempt to answer the question regarding the perception of users toward their physics-adapted avatars, there are several limitations to the current study that warrant further research on this topic. First, we put lots of consideration on how we could make independent conditions for human movement, as human movement is a kinetic chain. For example, the leg joint's position and rotation depend on the upper leg joint's movement. As such, fixing a particular joint would cause other interlocking joints to be fixed accordingly. Future work could include more human pose categories and motions with independent conditions. Second, we suggested a method to observe the remote space using the concept of "portal". We put a lot of thought into how to show the remote space, such as changing perspective, video streaming, or showing a post-video, and chose the "portal" that suited our purpose the most. In future studies, the effect of different methods of showing remote space could be explored. Third, we conducted user testing in a VR setting due to limitations in implementing an AR telepresence system, such as the small field of view(FoV) and the difficulty of showing the other space using a portal in low resolution. Conducting user testing in AR would allow users to have higher realism. Fourth, observing additional data such as the user's behavioral change or gaze, especially when a portal pops up, would allow deeper analysis. Unintentional gaze differences can show the user's consciousness toward the avatar [73].

In conclusion, we explored user preference and perceptions toward the avatar where synchronization between a user and an avatar causes the avatar to violate the laws of physics in remote space. Users reported feelings of less ownership and control over their avatar, a decreased level of intention and trust toward the system, and less preference when their pose category was modified. Physics conflict also had a negative impact on users' perceptions. Users recognized the hierarchy that occurred due to the eye level change, and generally preferred preserving the eye level of their local room. Taken together, our work shares insights into how users perceive their physics-adapted avatars in remote heterogeneous spaces.

CRediT authorship contribution statement

Taehei Kim: Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Hyeshim Kim:** Visualization, Software, Investigation. **Jeongmi Lee:** Writing – review & editing, Validation, Methodology. **Sung-Hee Lee:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.cag.2025.104185.

Data availability

Data will be made available on request.

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